BIODIVERSITY

Nature’s services and contributions

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Structure

1. Introduction of values
2. What is nature worth?
3. Calculation of values
4. Forest issues and threats
5. Nature based solutions

https://finlandnaturally.com/

1997-2018

What is nature worth?

...and who gets to decide?

US$33 Trillion

..a year. That is what nature does for us for free.

https://en.wikipedia.org/wiki/Biosphere_2

1.8X of the economy

One bee, two different persception

https://a-z-animals.com/animals/bumble-bee/

A service :

Bees moving pollen
Soil formation

≈ US$117 billion / year

Practice of care :

- Forests with nesting area for pollinators
- Totemic relationships, with obligation in both directions

Are the 2 papers opposites? ...not really

Similarity

- Both cover the same ground; food, water, regulation
- Admits their category overlap
- Want nature to count in decisions

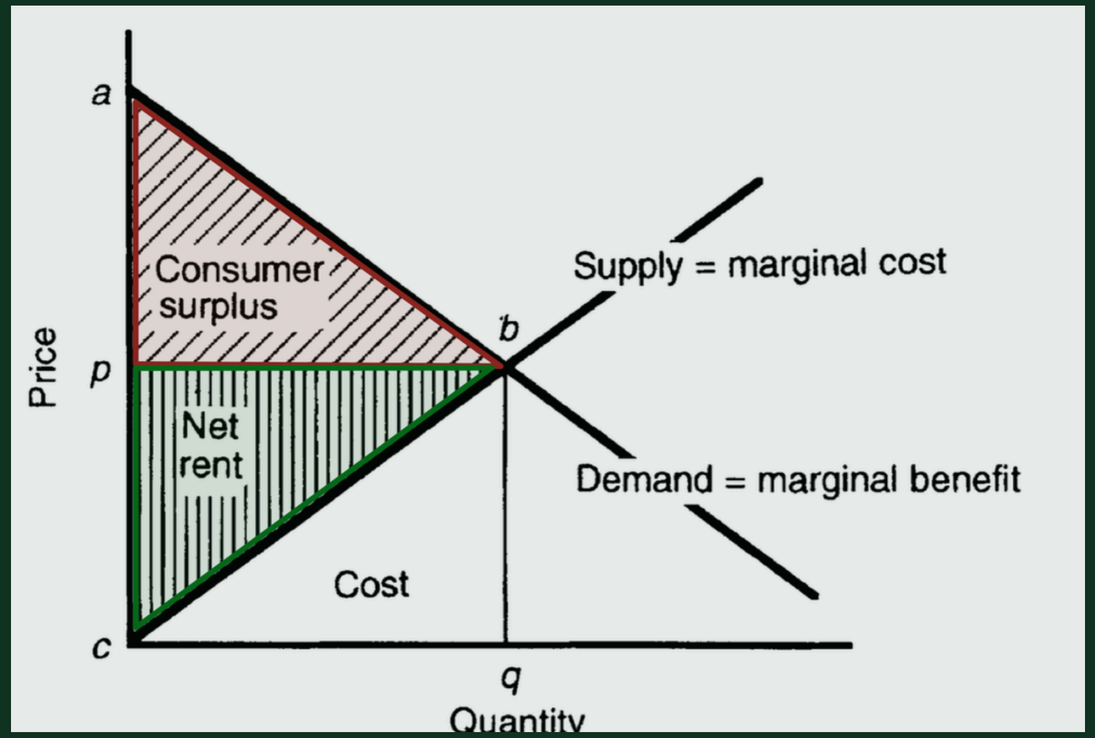
Differences

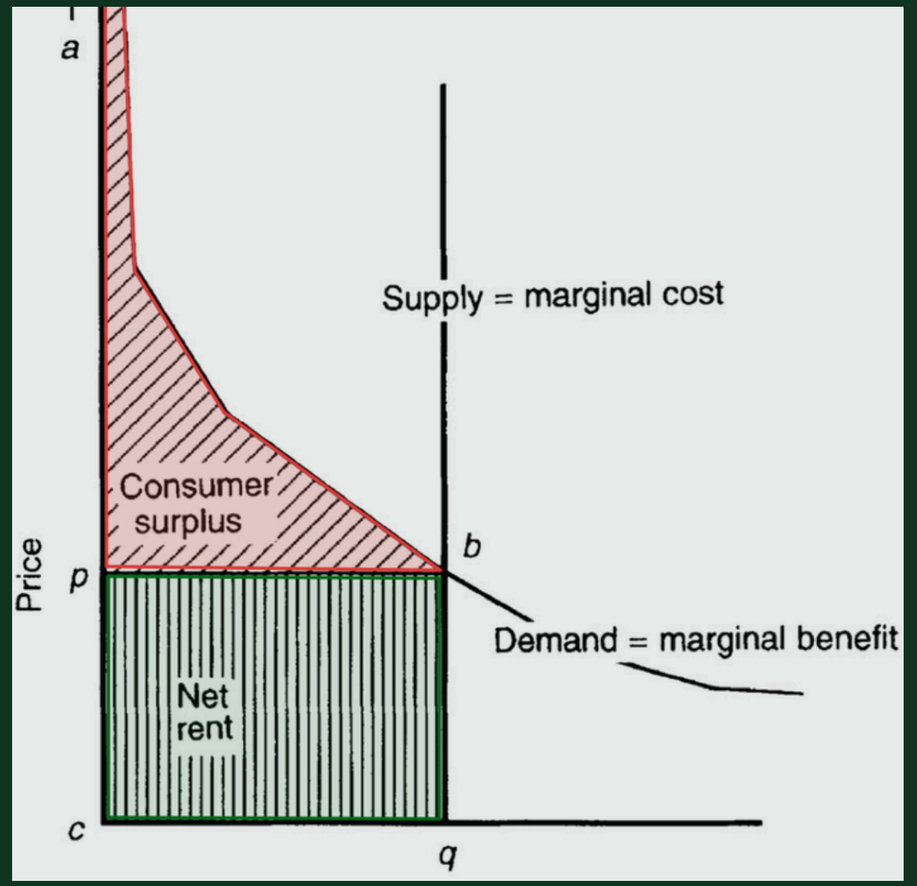
- How value is expressed
- 1997 speaks in dollar values, 2018 speaks of multiple ways of describing value

- **Tropical forest was US$2007 /hectare /year**

- **All the forest ≈ US$4.7 Trillion /year**

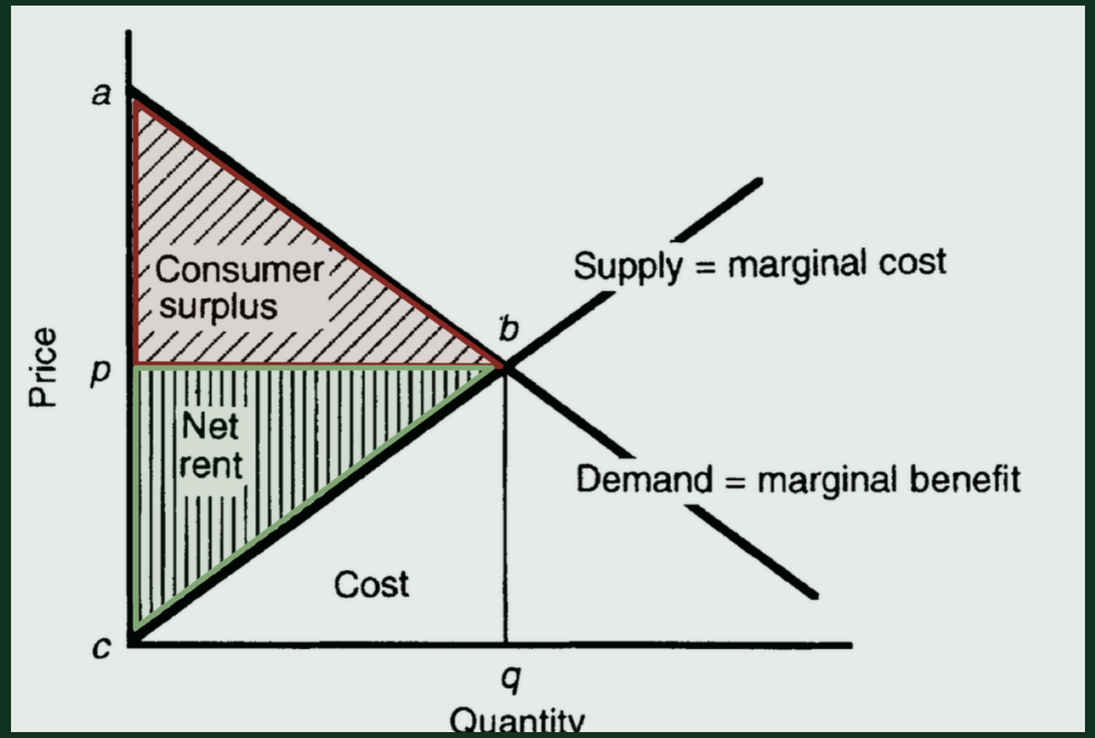
Calculation of values

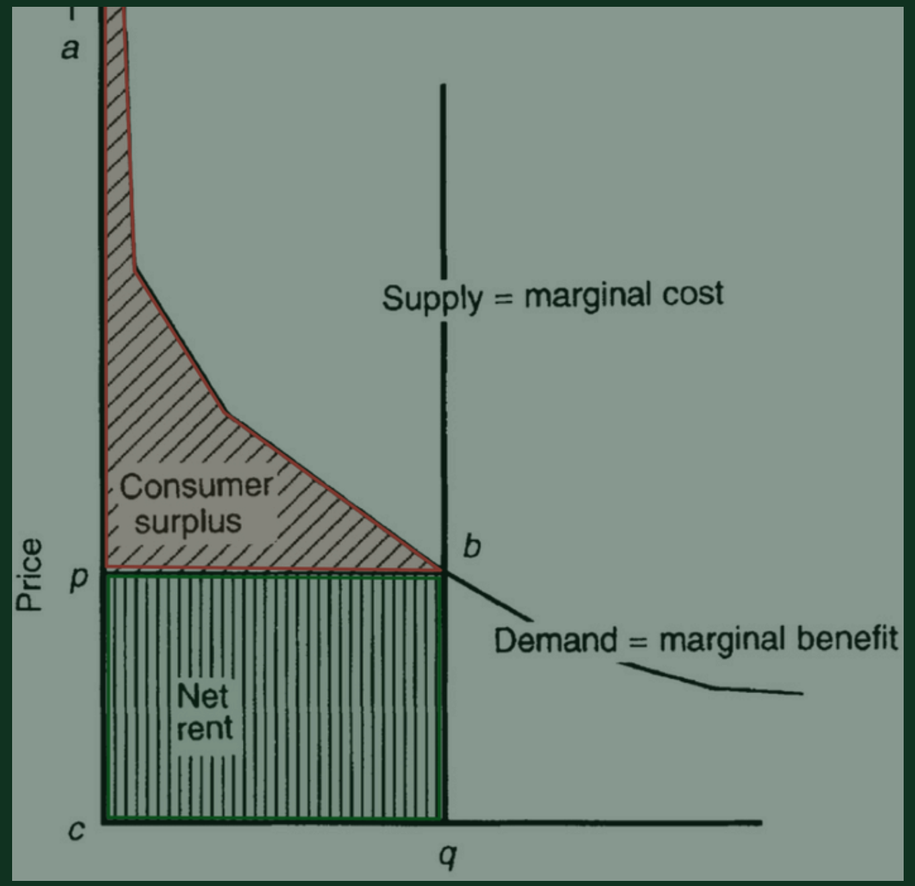




Costanza et al. 1997.pdf

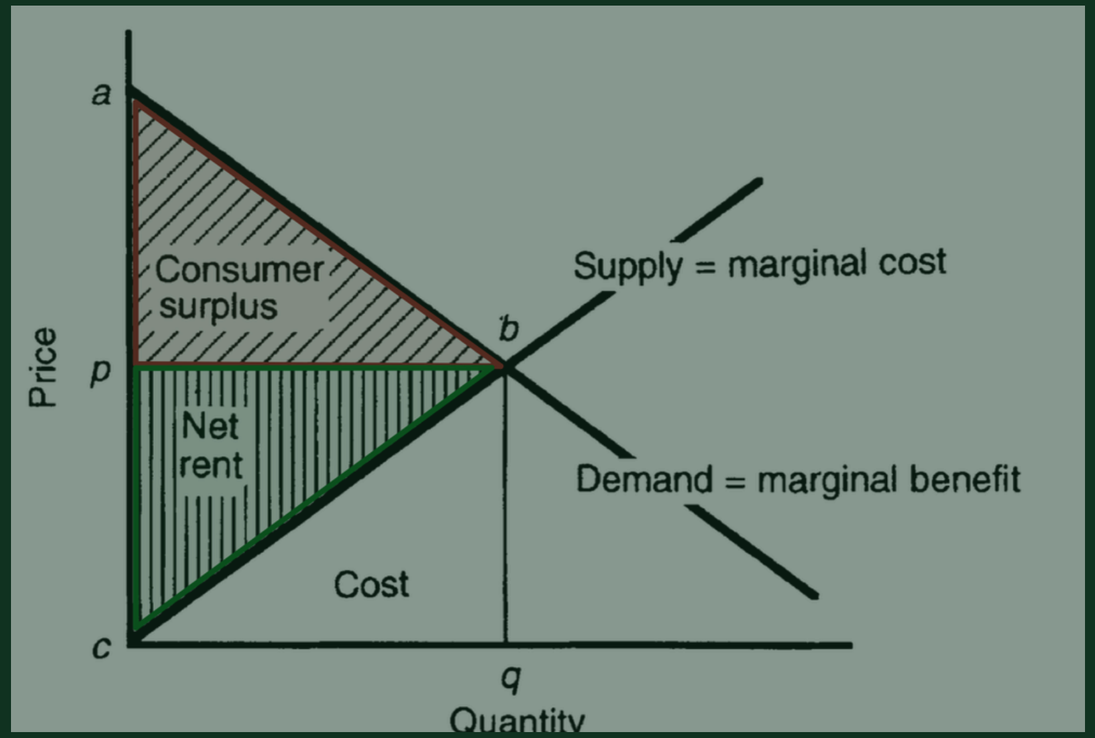
Regular chart

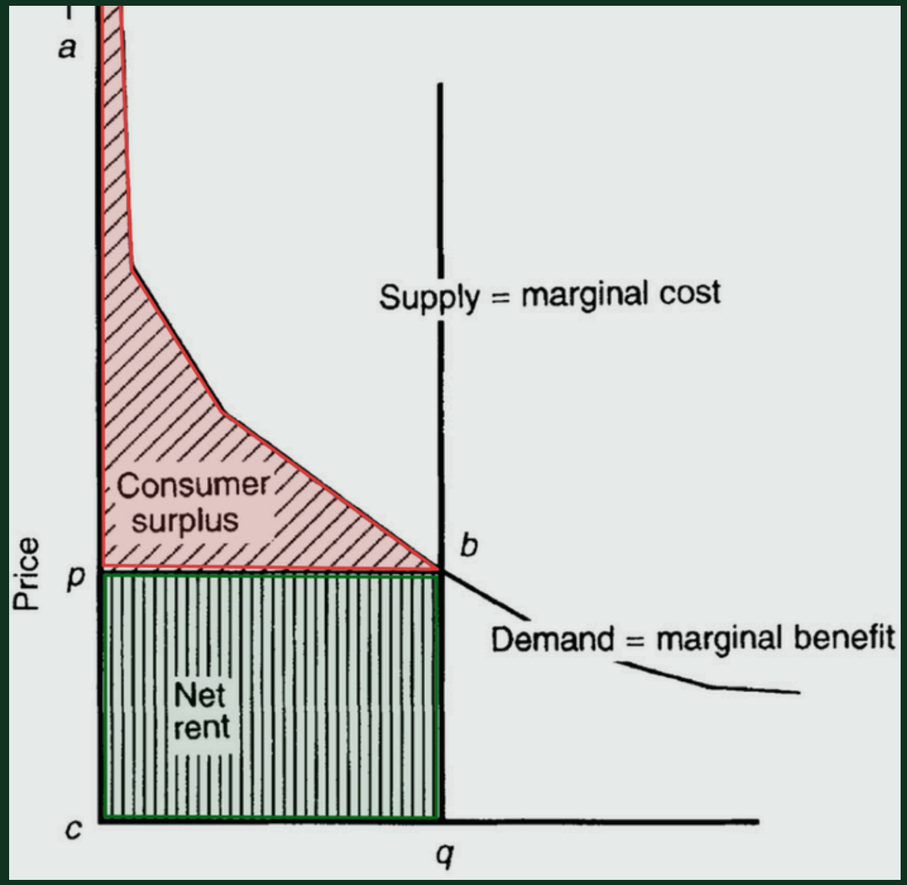




Costanza et al. 1997.pdf

Chart used for the calculations





Costanza et al. 1997.pdf

Table 2 Summary of average global value of annual ecosystem services

Biome

Area (ha x 10^8)

Ecosystem services

Biome	Area (ha x 10^8)	1 Gas regulation	2 Climate regulation	3 Disturbance regulation	4 Water regulation	5 Water supply	6 Erosion control	7 Soil formation	8 Nutrient cycling	9 Waste treatment	10 Pollination	11 Biological control	12 Habitat/ refugia	13 Food production	14 Raw materials	15 Genetic resources	16 Recreation	17 Cultural	Total value per ha ($ha^-1 yr^-1)	Total global flow value ($yr^-1 x 10^9)
Marine	36,302																		577	20,949
Open ocean	33,200	38							118			5		15	0			76	252	8,381
Coastal	3,102			88					3,677			38	8	93	4		82	62	4,052	12,568
Estuaries	180			567					21,100			78	131	521	25		381	29	22,832	4,110
Seagrass/ algae beds	200								19,002						2				19,004	3,801
Coral reefs	62			2,750						58		5	7	220	27		3,008	1	6,075	375
Shelf	2,660								1,431			39		68	2			70	1,610	4,283
Terrestrial	15,323																		804	12,319
Forest	4,855		141	7	7	3	96	10	361	87		2		43	138	86	66	2	969	4,706
Tropical	1,900		223	5	6	8	245	10	922	87				32	315	41	112	2	2,007	3,813
Temperate/boreal	2,955		88		0			10		87		4		50	25		36	2	302	894
Grass/rangelands	3,898	7	0		3		29	1		87	25	23		67		0	7		232	906
Wetlands	330	133		4,539	15	3,800				4,177			304	256	106		574	881	14,785	4,879
Tidal marsh/ mangroves	165			1,839						6,696			169	456	162		658		9,980	1648
Swamps/ floodplains	165	266		7,240	30	7,600				1,659			439	47	49		491	1,761	19,580	3,231
Lakes/rivers	200				5,445	2,117				665				41			230		8,498	1,700
Desert	1,925																			
Tundra	743																			
Ice/rock	1,640																			
Cropland	1,400										14	24		54					92	128
Urban	332																			
Total	51,625	1,341	684	1,779	1115	1,692	576	53	17,075	2,277	117	487	124	1,386	721	79	815	3,015		33,268

Costanza et al. 1997.pdf

Ecosystem services

Disturbance regulation

- Storm protection
- Flood control

Nutrient cycling

Biological control

Genetic recourses

Refugia

Ecosystem services

Disturbance regulation

Nutrient cycling

- Nitrogen fixation

Biological control

Genetic recourses

Refugia

Ecosystem services

Disturbance regulation

Nutrient cycling

Biological control

- Keystone predator control of prey species
- Reduction of herbivory by top predators

Genetic recourses

Refugia

Ecosystem services

Disturbance regulation

Nutrient cycling

Biological control

Refugia

Genetic recourses

- Medicine
- Products for material sciences

Ecosystem services

Disturbance regulation

Nutrient cycling

Biological control

Genetic recourses

Refugia

- Nurseries
- Habitat for migratory species

Forest issues and threats

www.nationalgeographic.com

www.plasticpollutioncoalition.org

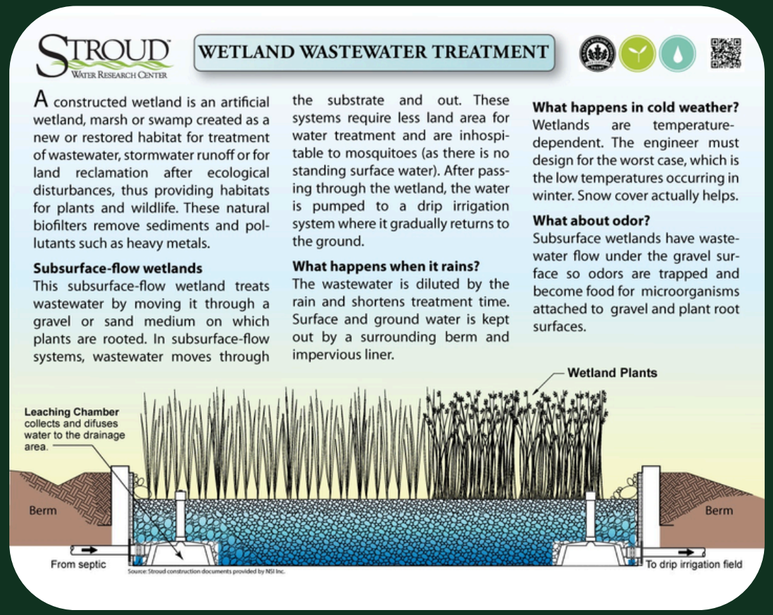
daily.jstor.org

Nature based solutions to aforementioned problems

Protect, manage or restore ecosystems to solve human problems, helping both people and biodiversity

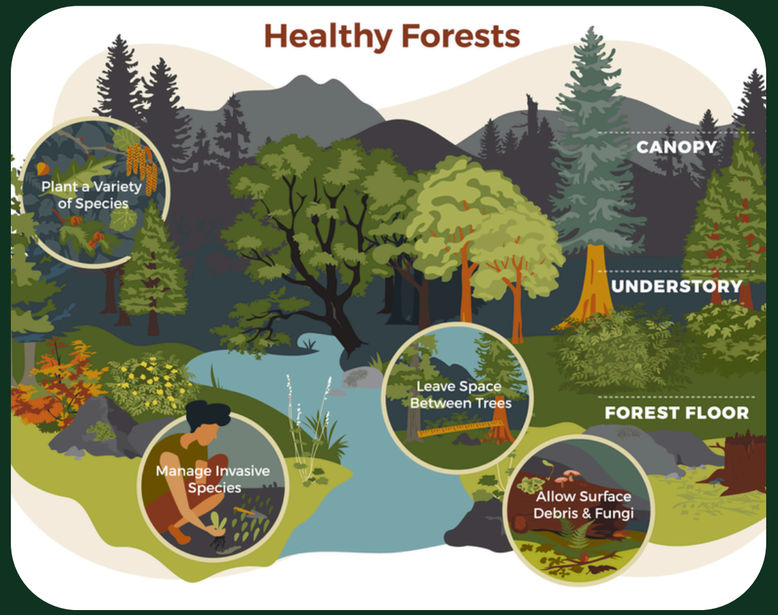
Reforestation
Restoring tree cover

Www.nature.org



Constructed wetlands
Natural water purification

Stroudcenter.org

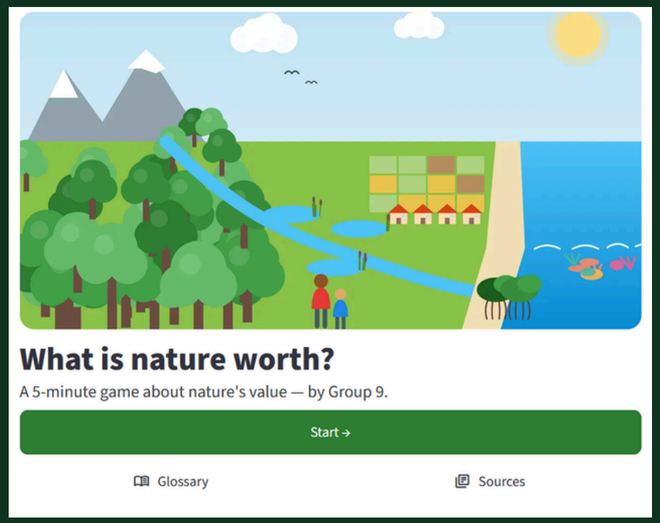


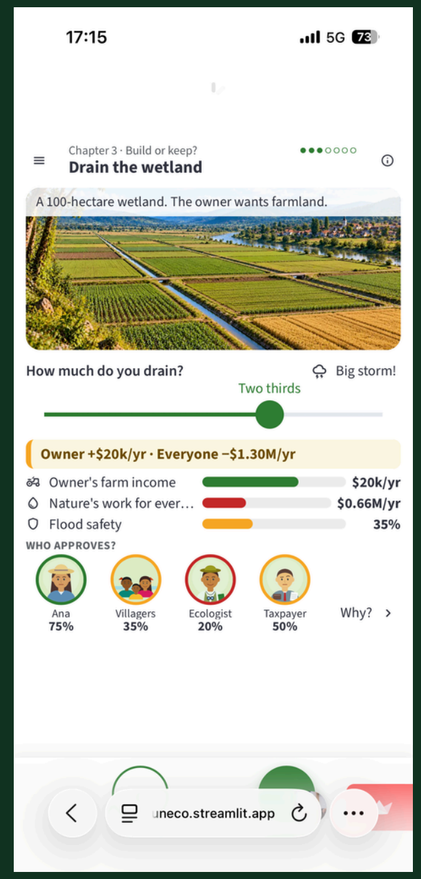
Forest management
Reducing wildfire risks

Tualatinswcd.org

...but nature-based solutions need space, time and local people.

What about our groups app





Play. Decide. Protect.

funeco.streamlit.app

github.com/zhanxdani/nature-value-explorer

Credentials

Julie DiNicola

- group coordination
- presentation structure
- values research
- forest issues research

Goda Šturaitė

- group co-coordinator
- nature’s services research
- Constanza’s calculations research
- presentation’s aesthetics

Munkhtulga Gantulga

- ecosystem services research (1997 vs 2018)
- examples and comparison of the two papers

Zhantai Dan

- Nature based solutions
- funeco mini-app
- group programmer

Sources:

- Diaz et al. 2018.pdf
- Costanza et al. 1997.pdf
- https://ecologyandsociety.org/vol30/iss4/art3/
- Threats | Forest health | Forest Information System of Europe https://share.google/KKoQN3cyzqAoDaiYs
- https://www.greenpeace.org/usa/forests/issues/
- What is forest biodiversity? | European Forest Institute https://share.google/d8hOW0owV4h4pel1L
- Agribusiness & Deforestation - Greenpeace - Greenpeace https://www.greenpeace.org/usa/forests/issues/agribusiness/
- https://www.tandfonline.com/doi/full/10.1080/26395916.2019.1669713
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- https://stroudcenter.org/wp-content/uploads/2016/08/wetlands_big.jpg
- https://tualatinswcd.org/wp-content/uploads/2021/11/052021-TSWCD-HealthyForests@2x.png
- https://github.com/zhanxdani/nature-value-explorer
- https://funeco.streamlit.app

Thank You for your attention